

Piyush Singh

AI ML Engineer | Data Science | Data Analyst

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EDUCATION

- ❖ Vellore Institute of Technology, B. Tech in Computer Science and Engineering (Artificial Intelligence and Machine Learning) CGPA: 8.73
- ❖ Vanasthali Public School CBSE–Class 12th • Scored 88.4% in CBSE Board Examinations
- ❖ Vanasthali Public School CBSE–Class 10th • Scored 96.4% in CBSE Board Examination

TECHNICAL SKILLS

Programming Languages: Python, C++

AI and ML skills: ML, DL, Neural Network, GenAI, LLM, Advance RAG, Ragas, Transformer (HuggingFace), NLP, Computer Vision AI ML

Framework and Tools: LangChain, LangGraph, LangSmith Monitoring, CrewAI, Ollama, MLflow, DVC, Docker, CI/CD, Scikit-learn, OpenCV, NLTK, Git, Docker, AWS, PowerBI

Backend and Database: API development using FastAPI and Flask, SQL, Redis, Postgres

Data Science: Python, Statistical Analysis & Modeling, ML, pandas, NumPy Data Visualization using PowerBI, PySpark, OpenSearch

PROJECTS

Water Potability Detection: MLOPS and CI/CD

https://github.com/PiyushVIT346/water_potability_MLOPS

- Trained RandomForest model to detect if the water is potable to drink. It uses data of mineral and salt in water having accuracy of 94.2%.
- Administered the model lifecycle for streamlined efficiency using MLflow and DagsHub, incorporating rigorous tracking, versioning, and a centralized registry, which improved model accuracy by a measurable 8.6%. and reducing data processing time by 30% and improving model training efficiency.
- Deployed FastAPI-based ML APIs with CI/CD automation via GitHub Actions, which reduced manual deployment.

ArXivIQ — Intelligent search & Q/A over research papers:

<https://github.com/PiyushVIT346/Research-Paper-RAG---production-graded>

- Help to ask Q/A for the latest research paper (ArXiv published) for any topic or category - CLI based project.
- Built a production-grade RAG system for arXiv research papers using FastAPI, Postgres (Neon), OpenSearch, and Gemini, implementing end-to-end ingestion, hybrid search (BM25 + KNN vector) in OpenSearch with custom index mappings, section-based chunking with overlap, and Jina AI embeddings for high-recall semantic retrieval, and LLM-based question answering with observability.
- Designed a resumable, fault-tolerant ingestion pipeline (arXiv API → PDF download → parsing → chunking → embedding → indexing) with rate limiting, caching, structured logging, and Prometheus metrics. Engineered an agentic retrieval pipeline with LLM-based guardrail scoring, query refinement, document relevance grading.
- Added Redis response caching and LangFuse tracing to the RAG pipeline, achieving sub-second latency on cached queries and full observability into guardrail, retrieval, and generation stages.

Trained Model for CT-Scan, X-Ray and Brain tumor

<https://github.com/PiyushVIT346/health-hackathon>

- Built a medical image classifier using CNNs and fine-tuned ResNet-50 to accurately distinguish X-rays, CT scans, and Brain MRIs across heterogeneous datasets, achieving 89% accuracy.
- Developed an NLP-powered disease prediction chatbot that processes user symptom inputs to identify probable conditions and suggest first-aid measures.
- Engineered a rule-based meal plan generator leveraging user health history, dietary preferences, and chronic conditions to deliver personalized nutrition recommendations.

EXPERIENCE

Intern and Trainee (Machine Learning and Data Science) at LearnFlu | Remote | November 2024 – January 2025

- Completed a 3-month internship focused on Python, machine learning, statistics, and data analysis with hands-on training in Pandas, NumPy, Scikit-learn, and Matplotlib.
- Applied practical experience in data preprocessing, feature engineering, model training, evaluation, and deployment across real-world business scenarios.
- Formulated ML solutions including Iranian churn prediction (Logistic Regression, RF, XGBoost), customer segmentation (RFM, KMeans, PCA, DBSCAN), and real estate price prediction using regression techniques.

Deep Learning Intern at UptoSkills | Remote | June 2026 – August 2026

- Worked on real-time project under Crowd Surveillance group, build a project to detect and classify the car manufacturing company for Indian Road and Jam test with precision of 91.2% and map-90 score 88.2%. Used computer vision to enhance the image. Used 2-layer of YOLO model that can reduce the latency and power.
- Promoted as Captain and working in management team. Enhance my communication and management skill.

RESEARCH PAPERS

- EMG Signal Disease Detection– Published in JETIR. <https://www.jetir.org/papers/JETIR2507378.pdf>

CERTIFICATION

- Microsoft DP-900: Azure Data Fundamentals | *June 2025*
- AWS AI Practitioner Certified | *12 May 2026*
- IBM GenAI Course | *June 2025*
- ServiceNow Virtual Internship program | *27 July 2026*
- LangChain, LangGraph and Ollama-RAG– Udemy | *January 2025*
- MCP: Build App with Claude LangChain Ollama | *November 2025*
- Intermediate Machine Learning– Kaggle | *July 2024*

Achievement and Honors

- Got 2nd Position (Domain wise: Data Analyst) in MahaKhumbh Hackathon.
- Selected as top 60 finalist in Health Hackathon 2025.
- Qualified till second - last round in NASSCOM GenAI hackathon. Build AI for Insurance claim just using phone call and most fascinating idea was to eliminate the useless question making it more accessible to both customer and vendor.
- Selected out of 80,000 students from India as mentee for graphs camp by Codeforce Master and learned 17+ advance problem-solving technique in graphs. Solved 1 out of 3 super hard problem on the topic Uber: Bus problem and got featured in Hall of fame.
- Participated in many hackathons: Google Agentic AI (H2Skill exchange hackathon), EY techathon 6.0, Flipkart E-Commerce, MarQing Minds, Elevate-24 Neural Network, HP power Quiz, Adobe India hackathon, TVS credit epic 7. Instead, we got no position, but got new experience that enhance our skills and motivate for further participations